

Adaptive Variable Resolution 2.5D Lidar Mapping

A defence ground vehicle's lidar sees **93% of what sticks up** and **15% of what it can fall into**.

One geometric test closes that gap — at a twentieth of the point budget.

DRDO

Smart Automation

Software

Team Raysense

Negative obstacles are invisible at *any* budget

Ditches, craters and washouts are detected by the **absence** of laser returns. A narrow trench is not entered by beams — it is stepped over. Measured on 40 full scans, every ray, no budget limit:

92.9%

of **positive** obstacles detected
things that stick up

11.5%

of **negative** obstacles detected
things you fall into

6.4%

of real hazards reported as **safe to drive over**
and this rises with more data

Spending more rays cannot fix it. A full scan already spends everything. More data makes the system more *confidently wrong*: it sees the trench rim, judges it flat, and calls it drivable.

A ditch leaves a gap that geometry cannot explain

Two adjacent beams that both return land a predictable distance apart on level ground. If the far one lands much further out, the surface between them dropped away.

The comparison must be against what geometry predicts **at that range** — ground spacing grows quadratically, so a fixed threshold flags flat ground seven times over.

$$\text{anomaly} = \text{measured gap} \div \text{predicted gap}(r)$$

SCENE	MAX RATIO	FLAGS
Flat ground	—	0
Rolling terrain	2.48	70
2.4 m × 1.8 m trench	7.28	299

Threshold **3.0**, chosen by sweep against the identical terrain with the ditches removed: it costs six points of recall against 2.0 and removes **89% of false flags**.

DETECTOR COST 43 ms/FRAME

The same formula that states the danger finds it.
The quadratic falloff is why ditches are missed — and, normalised, it is the detector.

BUDGET	ABSENCE TEST	OURS
2%	6.7%	74.9%
5%	9.7%	91.4%
100%	15.2%	99.8%

20× fewer points

5% budget with the detector finds **91%**.
A full scan without it finds **12%**.

What the geometry also tells you: how fast you may drive

A bump of height h at range r needs angular spacing h/r . A **ditch of width w needs $w \cdot h/r^2$** — quadratically tighter. A safety floor sized for bumps is comfortably met while ditches stay invisible.

$$\Delta\theta \leq w \cdot h_{\text{sensor}} / r^2$$

RANGE	NEEDED	OS1-64 HAS	MET?
8 m	1.611°	0.714°	yes
12 m	0.714°	0.714°	the limit
20 m	0.258°	0.714°	no
40 m	0.064°	0.714°	no

23 km/h

Maximum safe speed for a stock OS1-64 that must never drive into a 1 m ditch — and the sensor gives no indication that this is so.

Which lidar does this run on?

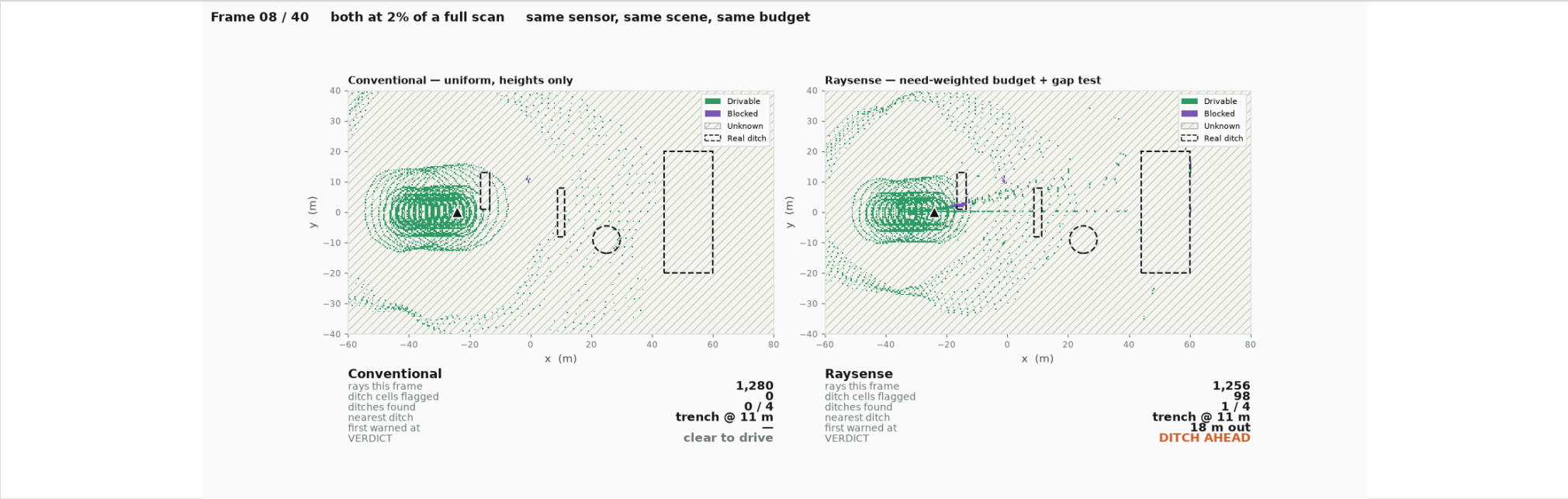
- **Steerable** (MEMS/OPA) — emits only budgeted rays; saves laser energy.
- **Fixed-pattern** (Velodyne, Ouster) — every ray fires; the budget selects what is processed. Saves bandwidth, memory and compute — *not* laser power.
- **Replay** — the evaluation harness.

What we deliberately do not build

- SLAM — we consume ego-motion, we do not estimate it
- Sensor fusion — lidar only, so the measurement is not confounded
- Path planning, dashboards, cloud

Python · NumPy · 103 tests · every figure generated from a committed CSV, none typed by hand.

Same sensor, same budget, opposite verdicts



1.88x

more warning distance at 2% budget
5.8 s to react, against 3.1 s

91%

of negative obstacles found
at 5% of a full scan

18 m → 0 m

we warn **18 m out**; the conventional
system has still said nothing at **11 m**

This field exists. We read it first.

Adaptive lidar sampling is established

Pittaluga et al., *Towards a MEMS-based Adaptive LIDAR*, NEC Labs (arXiv 2003.09545) · *Adaptive LiDAR Scanning: Harnessing Temporal Cues* (arXiv 2508.01562, 2025) · AEye iDAR / 4Sight, US 11675053 / 11782136 / 11860313 · *Fast Adaptive Scene Sampling for Single-Photon 3D Lidar*

So is negative-obstacle detection

Larson & Trivedi, *Lidar Based Off-road Negative Obstacle Detection*, DTIC ADA561293 · *Negative Obstacle Detection Using Sparse 3D LiDAR*, IEEE ROBIO 2024 · RELIS-3D, ICRA 2021

Honest limitations

- Crest occlusions still raise false flags; threshold 3.0 gives 73% precision, not 100%
- Results are on a controlled testbed with exact ground truth, not yet RELIS-3D
- Smart allocation does **not** beat plain decimation on whole-map recall — it wins only on warning distance

What is ours

Every published adaptive-lidar result targets urban driving, where negative obstacles barely exist. Every negative-obstacle result assumes dense uniform scanning.

The two fields have never been put in the same room.

We put them there, and found that adaptive sampling creates exactly the absences that hide a ditch — then derived a detector from the same geometry that causes the problem.

Team Raysense

Six members · internal hackathon nomination
Repository, findings and reproduction steps available.